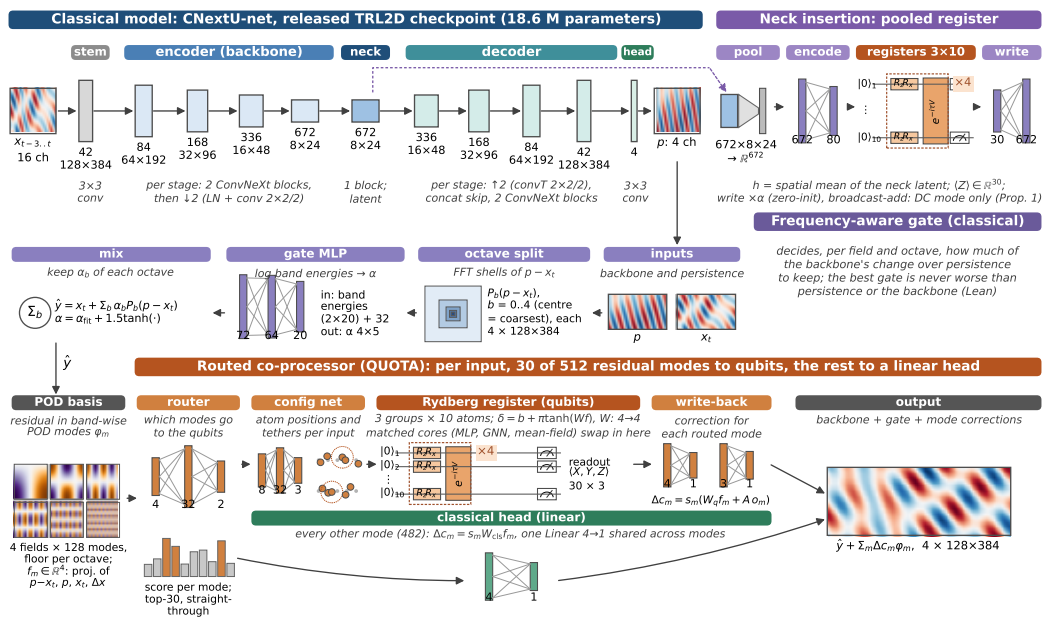

Classical model: CNextU-net, released TRL2D checkpoint (18.6 M parameters)
Neck insertion: pooled register
stem
encoder (backbone)
neck
decoder
head
pool
encode
registers 3×10
write
$x_{t-3..t}$
16 ch
42
128×384
84
64×192
168
32×96
336
16×48
672
8×24
672
8×24
336
16×48
168
32×96
84
64×192
42
128×384
4
p: 4 ch
672×8×24
→ $\mathbb{R}^{672}$
672
80
$|0\rangle_1$
$|0\rangle_{10}$
$R_x R_z$
$e^{-\gamma V}$
×4
30
672
3×3
conv
per stage: 2 ConvNeXt blocks,
then ↓2 (LN + conv 2×2/2)
1 block;
latent
per stage: ↑2 (convT 2×2/2),
concat skip, 2 ConvNeXt blocks
3×3
conv
h = spatial mean of the neck latent; $(Z) \in \mathbb{R}^{30}$;
write ×α (zero-init), broadcast-add: DC mode only (Prop. 1)
Frequency-aware gate (classical)
decides, per field and octave, how much of
the backbone's change over persistence
to keep; the best gate is never worse than
persistence or the backbone (Lean)
mix
keep α_b of each octave
Σ_b
$\hat{y} = x_t + \Sigma_b \alpha_b P_b(p - x_t)$
$\alpha = \alpha_{\text{filt}} + 1.5\tanh(\cdot)$
gate MLP
log-band energies → α
72
64
20
in: band
energies
(2×20) + 32
out: α 4×5
octave split
FFT shells of $p - x_t$
$P_b(p - x_t)$,
$b = 0..4$ (centre
= coarsest), each
4 × 128×384
inputs
backbone and persistence
p
x_t
$\hat{y}$
Routed co-processor (QUOTA): per input, 30 of 512 residual modes to qubits, the rest to a linear head
POD basis
residual in band-wise
POD modes φ_m
4 fields × 128 modes,
floor per octave:
$f_m \in \mathbb{R}^4$: proj. of
$p - x_t$, p, x_t, Δx
router
which modes go
to the qubits
4
32
2
score per mode;
top-30, straight-
through
config net
atom positions and
tethers per input
8
32
3
Rydberg register (qubits)
3 groups × 10 atoms; $\delta = b + \pi\tanh(Wf)$, W: 4→4
matched cores (MLP, GNN, mean-field) swap in here
$|0\rangle_1$
$|0\rangle_2$
$|0\rangle_{10}$
$R_x R_z$
$e^{-\gamma V}$
×4
readout
$\langle X, Y, Z \rangle$
30 × 3
write-back
correction for
each routed mode
4
1
3
1
$\Delta c_m = s_m(W_q f_m + A\, o_m)$
output
backbone + gate + mode corrections
classical head (linear)
every other mode (482): $\Delta c_m = s_m W_{\text{cls}} f_m$, one Linear 4→1 shared across modes
4
1
$\hat{y} + \Sigma_m \Delta c_m \varphi_m$, 4 × 128×384